



PEDIATRIC CARDIAC ARREST

Effective: January 1, 2026
Replaces: January 1, 2025

1. Patient Care Goals

- 1.1. Return of spontaneous circulation (ROSC)
- 1.2. Preservation of neurologic function
- 1.3. High-quality chest compressions/CPR with minimal interruption from recognition of cardiac arrest until confirmation of ROSC or field termination of resuscitative efforts
- 1.4. Recognition and treatment of reversible causes of cardiac arrest
- 1.5. For medical (non-traumatic) causes of cardiac arrest without obvious signs of death, 20 minutes of resuscitation efforts on-scene prior to making transport decision
- 1.6. Treatment consistent with patient's wishes as documented in Advanced Health Care Directives, POLST or DNR orders

2. BLS Treatment

- 2.1. If patient shows signs of obvious death (**Policy 600**), do not resuscitate
- 2.2. Confirm status of DNR / POLST / End of Life Option Act, if possible
 - 2.2.1. Do not delay care and/or CPR while confirmation is being made (**Policy 604**)
- 2.3. Address any areas of significant blood loss with hemorrhage control measures, regardless of any active bleeding or hemorrhage (**700-M17**)
 - 2.3.1. Apply tourniquet(s) proximal to any large wound, laceration, or amputation of the extremities, regardless of any active bleeding or hemorrhage
- 2.4. Apply **Spinal Motion Restriction (SMR)** if indicated (**700-M11**)
- 2.5. If traumatic cardiac arrest is suspected **initiate transport to appropriate receiving trauma center**, all remaining care to be completed en route to trauma center (**Policy 602**).
- 2.6. High quality uninterrupted CPR (**700-S01**)
- 2.7. Routine Medical Care – Pediatric (**700-S05**)
- 2.8. **Oropharyngeal airway (OPA)**
- 2.9. **BVM** – Ventilate once every three seconds (1:3), with supplemental oxygen
- 2.10. **Apply AED** and follow device instructions

3. ALS Treatment

- 3.1. Place patient on cardiac monitor and treat accordingly
 - 3.1.1. If the traumatic arrest patient is asystolic on initial contact, do not resuscitate
- 3.2. If cardiac arrest due to trauma, follow traumatic cardiac arrest algorithm in section 11
- 3.3. **Supraglottic airway device** (LMA Supreme)
 - 3.3.1. If Supraglottic airway attempt fails:
 - 3.3.1.1. **Oropharyngeal airway (OPA)**
- 3.4. **ETCO2** continuous numeric and waveform monitoring on every airway adjunct
- 3.5. **Vascular Access (IV) or Vascular Access (IO)**, per procedure (**700-M13**)

4. Ventricular Fibrillation and Pulseless Ventricular Tachycardia

- 4.1. **Defibrillation: 2 joules/kg, 4 joules/kg, 4 joules/kg**
 - 4.1.1. Starting with lowest energy setting (2 joules/kg)
 - 4.1.2. Each subsequent shock at 4 joules/kg



- 4.2. **Epinephrine (1:10,000) 0.01 mg/kg IV/IO**, repeat every 3-5 minutes for the duration of the arrest (Epinephrine is not indicated in traumatic cardiac arrest with hypovolemia from exsanguinating hemorrhage)
- 4.3. **Amiodarone 5 mg/kg IV/IO**, single dose only
- 4.4. **BASE CONTACT**: Consult for further instruction on Amiodarone dosages, if there is a successful conversion to a sustained pulsatile rhythm or (ROSC)

5. Asystole (Non-Traumatic)

- 5.1. **Epinephrine (1:10,000) 0.01 mg/kg IV/IO**, repeat every 3-5 minutes for the duration of the arrest
- 5.2. Provider may consider termination of resuscitative efforts after a total of at least twenty (20) minutes of EMS provider resuscitation if:
 - 5.2.1. Arrest was not witnessed by the EMS provider
 - 5.2.2. No return of spontaneous circulation (ROSC) prior to transport

6. Pulseless Electrical Activity (PEA)

- 6.1. Identify and treat any reversible causes:
 - 6.1.1. **Hypovolemia**: Reassess any hemorrhage control interventions to ensure they are adequately addressing blood loss and reapply if necessary **(700-M17)**
Consider a **20 ml/kg Fluid bolus**, repeat once if needed **(700-P10)**
 - 6.1.2. **Hypoxia**: Ensure that the patient is adequately ventilated, utilizing an airway adjunct and bag valve mask with a supplemental oxygen supply
 - 6.1.2.1. Ensure proper chest rise and fall
 - 6.1.2.2. Reassess any sucking chest wounds or flail segment interventions
 - 6.1.3. **Hyperkalemia**: Peaked T-waves, with possible widening of the QRS complex
 - 6.1.3.1. Consider **Calcium Chloride 10 mg/kg IV/IO**, max dose 1gm
 - 6.1.3.2. Flush the IV tubing well between injections
 - 6.1.3.3. Consider **Sodium Bicarbonate 1 mEq/kg IV /IO**, max dose 50mEq
 - 6.1.4. **Tension Pneumothorax**: If tension pneumothorax is suspected or the patient has a traumatic injury to the chest, perform bilateral pleural decompression if not already completed. **(700-M02)**
- 6.2. **Epinephrine (1:10,000) 0.01 mg/kg IV / IO**, may repeat every 3-5 minutes for the duration of the arrest (Epinephrine is not indicated in traumatic cardiac arrest with hypovolemia from exsanguinating hemorrhage)
- 6.3. Treat any rhythm changes according to correct treatment protocol.
 - 6.3.1. If the PEA changes to asystole the provider may follow the criteria in section 5

7. Hypothermic Cardiac Arrest

- 7.1. Assess pulse for 45 seconds
- 7.2. If no pulse is present, **Start CPR**
- 7.3. If defibrillation is indicated, limit to one (1) shock until patient is warm
- 7.4. If patient presents with dysrhythmias, treat as appropriate but withhold IV medications until temperature rises above 86°F
- 7.5. Consider rewarming measures **(700-P09)**
 - 7.5.1. Patients that are hypothermic can be unresponsive to pharmaceutical therapy and electrical therapy and should be transported with CPR and warming measures



8. Ventricular Assist Device (VAD) Cardiac Arrest

- 8.1. High quality uninterrupted CPR (**700-S01**) may be provided if:
 - 8.1.1. Patient is unresponsive, apneic and there is a device failure alarm with no rotor hum upon auscultation
 - 8.1.2. **Mechanical CPR devices are contraindicated**
- 8.2. If there is presence of rotor hum with no failure alarm, continue with airway management, do not perform chest compressions (**700-M01**)
- 8.3. Defibrillation(s) by manual defibrillator or AED may only be delivered if the patient is unresponsive
- 8.4. Any pediatric VAD patient in cardiac arrest, that does not meet obvious death criteria (**Policy 600**) shall be transported to:
 - 8.4.1. Stanford Hospital
- 8.5. Treat the cardiac arrest VAD patient with the guidelines found in the Ventricular Assist Device protocol (**700-S11**)
- 8.6. If further guidance is required during patient care, make **BASE CONTACT**

9. Pregnant Cardiac Arrest

- 9.1. Any pregnant patient beyond 20 weeks gestation (uterine fundus palpated at or above the umbilicus) in cardiac arrest, that does not meet obvious death criteria (**Policy 600**) shall be transported to:
 - 9.1.1. Either Stanford Hospital or Santa Clara Valley Medical Center (for potential resuscitative hysterotomy)

10. Drowning or Mechanical Asphyxiation

- 10.1. Any drowning or mechanical asphyxiation patient, in cardiac arrest, with suspected head or spinal injury shall be transported to a Pediatric Trauma Center (**Policy 602**)
- 10.2. If head or spinal injury is not suspected, transport to a Pediatric Receiving Center (**Policy 602**)

11. Traumatic Cardiac Arrest

- 11.1. **Initiate Immediate Transport to appropriate receiving Trauma Center**, all care to be completed en route to pediatric trauma center (**Policy 602**).
- 11.2. Place supraglottic airway (LMA Supreme) if unable to oxygenate/ventilate with BLS airway
- 11.3. Epinephrine is not indicated in traumatic cardiac arrest with hypovolemia from exsanguination; otherwise epinephrine can be used in traumatic cardiac arrest
- 11.4. Treat reversible causes of traumatic cardiac arrest
 - 11.4.1. Administer **20 ml/kg Normal Saline**, may repeat once, up to 40 ml/kg or 1 L
 - 11.4.2. Control hemorrhage per protocol (**700-M17**)
 - 11.4.3. Perform bilateral pleural decompression if evidence of chest trauma (**700-M02**)

12. Return of Spontaneous Circulation (ROSC)

- 12.1. If any Return of Spontaneous Circulation (ROSC) occurs,
 - 12.1.1. If systolic blood pressure is less than ninety (90) mmHg (age 10 and older), or less than 70 mmHg + 2x age years (age 0-9) treat according to (**700-P10**) Shock
 - 12.1.2. Continue ventilations at a rate and volume to keep ETCO₂ between 30-40 mmHg
 - 12.1.3. If wide complex tachycardia, perform **synchronized cardioversion** (**700-P14**)
 - 12.1.4. **Blood Glucose Level**, readings of less than 60mg/dl require interventions



- 12.1.5. Transport to closest Advanced Pediatric Receiving Center, unless a traumatic cardiac arrest

13. Pertinent Assessment Findings

- 13.1. Obvious signs of death (**Policy 600**)
- 13.2. Evidence of STEMI after ROSC
- 13.3. Reversible causes of cardiac arrest
 - 13.3.1. Hypoxia
 - 13.3.2. Hypovolemia
 - 13.3.3. Hypothermia
 - 13.3.4. Hypo-/hyperkalemia
 - 13.3.5. Tension Pneumothorax
 - 13.3.6. Toxins (e.g. tricyclic antidepressants, beta blockers, opioids)
- 13.4. ETCO₂ lower than 10mmHg despite high quality CPR

14. Key Documentation Elements

- 14.1. Resuscitation attempted and all interventions performed
- 14.2. Arrest witnessed
- 14.3. Location of arrest
- 14.4. First monitored ECG rhythm
- 14.5. CPR prior to EMS arrival
- 14.6. Outcome upon arrival at hospital
- 14.7. Any ROSC
- 14.8. Presumed cardiac arrest etiology



15. Treatment Flow Charts



